DESIGN AND IMPLEMENTATION OF ASSETS MANAGEMENT SYSTEM

BY

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NSUK/PT1/CMP/0048/18/19

BSC. COMPUTER SCIENCE

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BEING A PROJECT SUBMITTED TO DEPARTMENT OF COMPUTER SCIENCE, NASARAWA STATE UNIVERSITY KEFFI, IN PARTIAL FULFILLMENTS OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR DEGREE IN COMPUTER SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF NATURAL AND APPLIED SCIENCES

NASARAWA STATE UNIVERSITY, KEFFI, NIGERIA

DECLARATION

I hereby declare that this project, Design and Implementation of Assets Management System, has been written by me and it is a report of my research work. It has not been presented in any previous application for the BSc. Computer Science. All quotations are indicated and sources of information specifically acknowledged by means of references.

SULEIMAN EZEKIEL

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CERTIFICATION

This project, Design and Implementation of Assets Management System meets the regulations governing the award of Bachelor of Science (BSc.) Degree in Computer Science in the Faculty of Natural and Applied Sciences, Nasarawa State University, Keffi, and is approved for its contribution to knowledge.

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DEDICATION

This work is dedicated to my family, whose love, prayers, and unwavering support have been the foundation of my strength. It is also dedicated to all who believe in the pursuit of knowledge as a tool for growth and transformation.

ACKNOWLEDGEMENTS

I am deeply grateful to my supervisor, Dr. M. Olalere, for his patient guidance, constructive feedback, and constant encouragement throughout the course of this work. His expertise and commitment to academic excellence provided the direction needed to bring this project to fruition.

I also acknowledge with appreciation the contributions of my lecturers and colleagues in the Department of Computer Science. Their knowledge exchange, discussions, and support created a vibrant academic environment that made this journey both rewarding and memorable. I equally extend my gratitude to my office for granting me the opportunity and flexibility to pursue this postgraduate program, which made the experience fruitful and fulfilling.

Finally, I owe profound thanks to my family for their sacrifices, encouragement, and unwavering support. Their prayers and motivation sustained me through the challenges of this academic pursuit, and I remain indebted to them for making this dream a reality.

ABSTRACT

The purpose of this study was to design and implement a secure, lightweight, and multi-tenant Asset Management System (*Kite Assets*) capable of improving accountability and efficiency in organizations. The research was motivated by the persistent challenges of manual methods such as duplication, errors, and lack of security and the prohibitive cost and complexity of many commercial AMS solutions. Adopting a system development research design guided by the Iterative Software Development Life Cycle (SDLC), the system was implemented using Next.js, TypeScript, Prisma ORM, and Tailwind CSS, with Role-Based Access Control (RBAC) ensuring structured user privileges. The findings revealed that the system successfully automated asset lifecycle management, improved traceability through QR code integration, enhanced usability with theme customization, and provided scalable reporting features. Conceptually, the study concludes that an affordable, role-secured AMS can bridge the gap between inefficient manual approaches and costly enterprise platforms, contributing a practical and adaptable tool for SMEs and educational institutions.

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CHAPTER ONE
INTRODUCTION

1.1 Background to the Study

In the digital economy, asset management has become a critical factor in ensuring accountability and optimizing organizational resources. Assets such as computers, vehicles, office equipment, and software licenses represent significant capital investments that sustain business activities (Alhassan & Emmanuel, 2021). When properly managed, they enhance productivity, reduce risks, and support decision-making.

However, in many organizations, particularly within developing economies, management practices still depend on handwritten records and spreadsheets, which often result in inefficiencies, inaccurate data, and difficulties during audits (Okoro & Adepoju, 2022).

Over the last decade, the emergence of automated Asset Management Systems (AMS) has improved asset tracking across the lifecycle from acquisition to disposal through centralized and accessible platforms (Kumar & Singh, 2021). These systems commonly include functions such as real-time registration, maintenance scheduling, and reporting. Despite their benefits, adoption is limited by high cost, technical complexity, and lack of adaptation to local needs (Eze & Chukwu, 2023).

This study therefore proposes a web-based AMS that is efficient, cost-effective, and user-friendly, aimed at improving accuracy, accountability, and decision-making in asset management.

1.2 Statement of the Problem

Manual and outdated approaches to asset management remain common, especially in smaller organizations. Methods such as handwritten records, spreadsheets, and unlinked databases are prone to errors, redundancy, and data loss, resulting in misplaced assets, poor accountability, and inefficiency (Okoro & Adepoju, 2022).

Although commercial AMS platforms exist, they are often expensive, complex, and designed for large enterprises, making them unsuitable for organizations with limited ICT infrastructure (Eze & Chukwu, 2023). The absence of integrated features such as real-time updates and automated maintenance further reduces their effectiveness. Without accurate data, organizations cannot optimize resources or detect anomalies such as theft or unauthorized transfers (Afolayan et al., 2021).

This study addresses these challenges by developing a centralized, cost-effective AMS that automates asset registration, tracking, and lifecycle management while remaining adaptable and user-friendly.

1.3 Objectives of the Study

Aim:

The aim of this study is to design, develop, and test a cost-effective Asset Management System that enhances efficiency, accountability, and accuracy in organizational asset tracking.

The objectives are:

1. To design a centralized digital platform for asset management.

2. To develop and implement core functionalities such as asset registration, allocation, tracking, maintenance, and disposal.
3. To test the system to ensure reliability, usability, and effectiveness.

1.4 Significance of the Study

This study has operational, financial, strategic, and academic significance.

From an operational perspective, the system will streamline workflows, minimize human error, and improve the accuracy of asset records, making information retrieval faster and more reliable (Okoro & Adepoju, 2022).

Financially, the system will reduce losses associated with theft, misplacement, and unrecorded disposals. It will also provide depreciation insights to support budgeting and prevent unnecessary purchases (Afolayan et al., 2021).

Strategically, the AMS offers a scalable platform aligned with organizational growth. Its real-time reporting capabilities will strengthen accountability and compliance with audit requirements (Eze & Chukwu, 2023).

Academically, the study will contribute to the body of knowledge on affordable and context-specific AMS solutions for developing regions, providing a reference framework for future research.

1.5 Scope of the Study

This study covers the design, development, and implementation of a web-based AMS for managing both physical and digital assets. The system will provide a centralized platform where administrators and authorized users can register, track, allocate, maintain, and dispose of assets efficiently.

The system will include secure, role-based authentication, automated updates, and reporting functions to support auditing, budgeting, and decision-making. It will be deployed as a web application accessible through standard internet browsers.

The study does not extend to mobile application development, third-party integrations, or predictive analytics. Physical tagging technologies such as RFID or barcodes are also excluded from this initial version, though the system will be designed to support them in future upgrades. Testing will be conducted using simulated datasets that mirror real asset management scenarios.

CHAPTER TWO
LITERATURE REVIEW

2.1 Conceptual Framework

The conceptual framework provides the foundation for understanding how an Asset Management System (AMS) operates within an organizational setting. It highlights the transition from traditional methods of managing resources to technology-driven approaches that integrate automation, centralization, and security. The framework also situates the study within contemporary ICT practices, emphasizing the importance of Role-Based Access Control (RBAC) in ensuring data integrity. Together, these concepts define the basis upon which the proposed system is designed.

2.1.1 Asset Management System

Asset Management Systems are designed to register, track, and monitor resources throughout their lifecycle. Recent studies stress that AMS play a central role in improving efficiency by reducing redundancy, preventing misplacement of assets, and enabling real-time decision-making (Chatziamanetoglou & Rantos, 2023). Unlike manual or spreadsheet-based methods, AMS provide integrated modules that support acquisition, allocation, monitoring, maintenance, and disposal of both physical and digital assets.

In higher education and industrial environments, for instance, AMS have been deployed to manage IT equipment, laboratory tools, and infrastructure, ensuring compliance with audit requirements while optimizing utilization (Kormpakis, Kapsalis & Alexakis, 2023). This demonstrates the growing importance of AMS as essential tools for organizational accountability.

2.1.2 Role of Information and Communication Technology (ICT)

The adoption of ICT has reshaped asset management by enabling automation, scalability, and enhanced data accuracy. ICT-enabled AMS allow data to be stored in centralized databases that multiple users can access remotely and securely. Such systems incorporate dashboards, analytics, and reporting features, which provide managers with insights into usage trends and depreciation rates (Nyame & Qin, 2020).

In sectors such as energy and healthcare, ICT-driven AMS have been instrumental in predictive maintenance, reducing downtime, and extending the lifecycle of expensive resources (Kormpakis et al., 2023). This demonstrates that AMS are no longer passive databases but active decision-support systems embedded in organizational operations.

2.1.3 Security and Role-Based Access Control (RBAC)

As organizations grow, securing asset information becomes increasingly important. RBAC has emerged as the most widely used access control mechanism for AMS. By linking permissions to roles rather than individual users, RBAC simplifies administration while ensuring only authorized personnel perform critical operations (Mandru, 2019).

Recent improvements in RBAC models integrate contextual features such as time and location of access, creating hybrid models that blend RBAC with Attribute-Based Access Control (ABAC) (Chatziamanetoglou & Rantos, 2023). In multi-tenant systems, RBAC not only prevents unauthorized access but also supports data segregation, ensuring each organization maintains ownership of its resources.

2.1.4 Asset Lifecycle Model

The asset lifecycle underpins the conceptual framework of AMS. Assets typically progress through the stages of acquisition, allocation, usage/tracking, maintenance, and disposal. Each stage is interconnected, and poor management at one point often has consequences for the others. For example, delays in updating maintenance records can lead to premature asset failure or misuse (Nyame & Qin, 2020).

As shown in *Figure 2.1*, the lifecycle is supported by a centralized ICT-enabled database that consolidates asset records, while RBAC provides an overlay of security across all stages. This dual structure ensures that asset information remains accurate and accessible to authorized users, thus promoting transparency and operational efficiency.

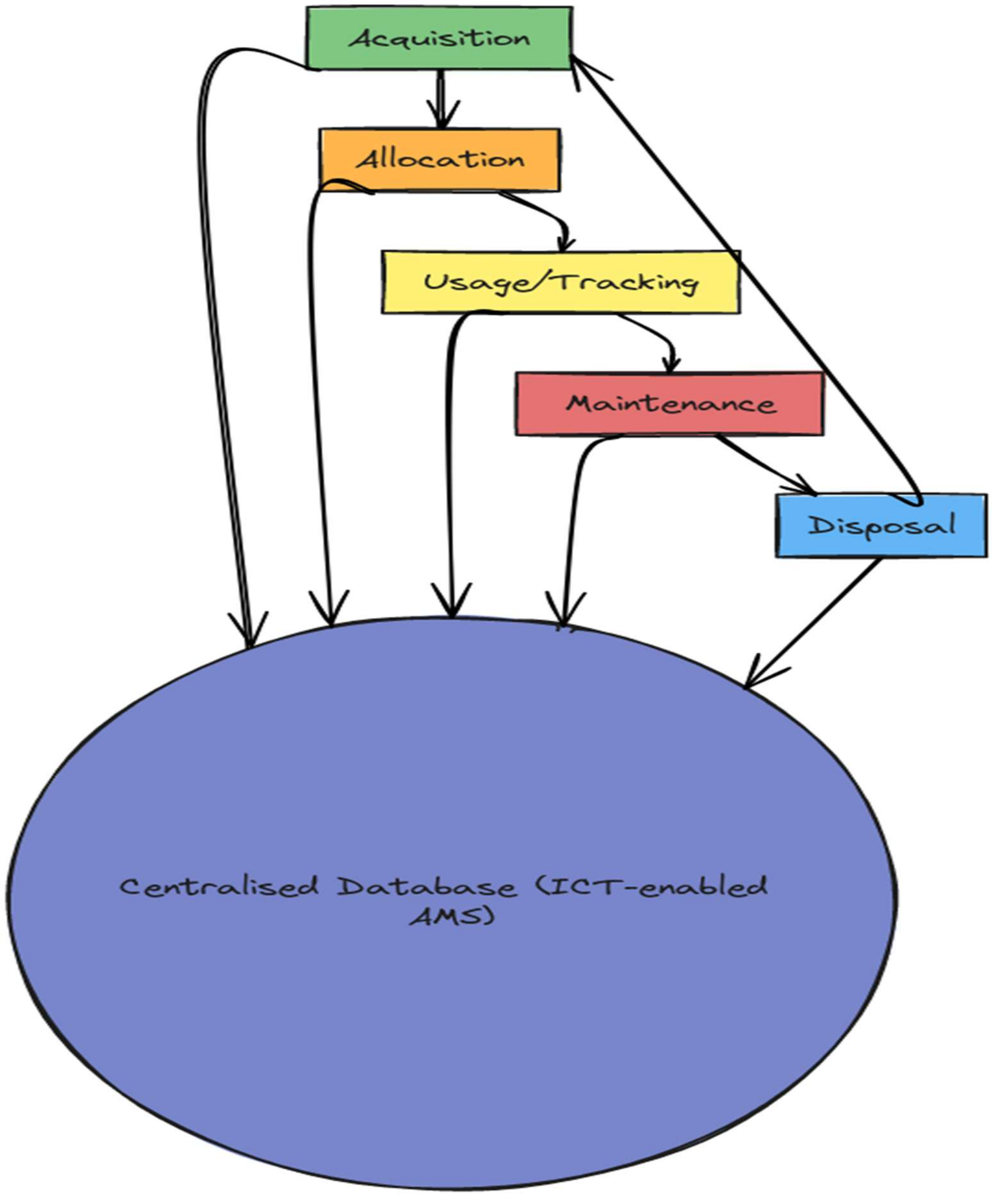

Figure 2:1 Asset Management System Lifecycle Model

This illustrates the cyclical flow of assets through acquisition, allocation, usage, maintenance, and disposal, all integrated into a central database governed by RBAC security.

2.2 Theoretical Framework

The theoretical foundation of this study is grounded in access control models, particularly the Role-Based Access Control (RBAC) paradigm. Asset Management Systems require strict control over who can view, modify, and dispose of assets, making RBAC central to the system's design. The framework also acknowledges contemporary extensions of RBAC, such as task-role and hybrid models, which provide flexibility and scalability for modern ICT environments.

2.2.1 Role-Based Access Control (RBAC) Theory

The RBAC model assigns permissions to roles rather than individuals. Users are then linked to roles according to their responsibilities within the organization. This reduces administrative overhead while ensuring security and accountability. For instance, staff may only register or update asset usage, managers may approve allocations, while administrators manage catalogues and user accounts.

Nyame and Qin (2020) note that RBAC improves policy consistency by enforcing uniform rules across similar roles, reducing the risks of errors and privilege misuse. Furthermore, RBAC scales effectively in multi-tenant environments, where each organization maintains its own hierarchy of roles and permissions.

2.2.2 Extensions of RBAC Models

Contemporary studies have highlighted the limitations of static RBAC, particularly in dynamic environments where access rights may depend on context. To address this, hybrid models have been developed:

I. **Task–Role-Based Access Control (TRBAC):** Assigns permissions not only by role but also by specific tasks or workflows. This is suitable for environments with process-driven operations (Mandru, 2019).

II. **RBAC with Smart Contracts:** Blockchain-enabled RBAC ensures transparent and tamper-proof access control in distributed AMS, enhancing trust in multi-organization platforms (Chatziamanetoglou & Rantos, 2023).

III. **Hybrid RBAC–ABAC Models:** Combine role-based structures with contextual attributes such as time, device, or location, offering greater flexibility for cloud-hosted AMS (Kormpakis, Kapsalis & Alexakis, 2023).

These extensions demonstrate how RBAC continues to evolve, keeping pace with security requirements in modern ICT systems.

2.2.3 Relevance of RBAC to Asset Management Systems

In AMS, RBAC directly contributes to operational accountability. By ensuring that only authorized users perform sensitive functions such as deleting assets or reallocating resources, RBAC protects the integrity of organizational data. It also reduces insider threats by preventing privilege escalation, a common issue in systems with poorly managed access rights.

The theoretical grounding of this study, therefore, rests on the principle that RBAC not only secures data but also supports organizational efficiency by aligning permissions with roles. This makes it particularly suitable for the design of multi-tenant AMS, such as the system proposed in this research.

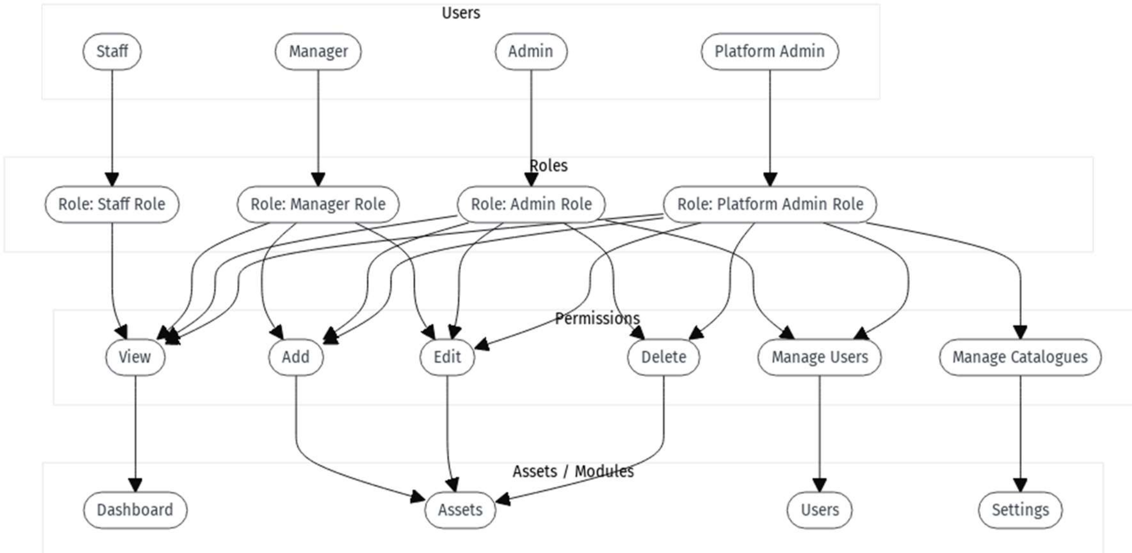

Figure 2:2 Role-Based Access Control Model

This demonstrate how RBAC theory translates into the proposed AMS.

2.3 Empirical Review

The empirical review focuses on how Asset Management Systems (AMS) and Role-Based Access Control (RBAC) have been applied across different organizational sectors. By examining evidence from education, healthcare, industrial enterprises, and small and medium-sized enterprises (SMEs), this section identifies both the benefits and the limitations of existing approaches.

2.3.1 Asset Management in Educational Institutions

Educational organizations increasingly rely on AMS to manage ICT equipment, laboratory resources, and facilities. The adoption of centralized digital platforms has improved audit compliance and reduced the risk of asset misplacement. For instance, recent findings show that digital AMS enhanced transparency and reduced redundancy in university inventory systems by consolidating records into a single database (Nyame & Qin, 2020). Similarly, studies in technical training institutions indicate that integrating AMS with analytics improved utilization and supported more accurate budget planning (Kormpakis, Kapsalis & Alexakis, 2023).

Despite these benefits, barriers remain. Many institutions in developing contexts face challenges such as poor ICT infrastructure and limited funding, which restrict the adoption of commercial AMS. This gap highlights the importance of web-based solutions that are lightweight, cost-effective, and adaptable to institutional needs.

2.3.2 Asset Management in Healthcare

Healthcare organizations require robust systems to track medical devices, consumables, and digital health records. AMS integrated with RBAC has proven effective in protecting sensitive data while ensuring operational efficiency. Chatterjee, Pitroda and Parmar (2020) demonstrated that dynamic RBAC models in healthcare settings allowed flexible role assignments, improving access without compromising security. More recently, blockchain-enabled AMS frameworks have been tested in hospital environments to improve data integrity and prevent unauthorized modifications of patient-related assets (Zaidi, Usman, Aftab & Aljuaid, 2023).

However, interoperability across departments remains a challenge. Many hospitals struggle to integrate AMS with existing legacy systems, leading to fragmented data and inefficiencies.

2.3.3 Asset Management in Industrial and Energy Sectors

In industry, AMS supports predictive maintenance and lifecycle optimization of machinery. Case studies in energy firms report that AMS linked with RBAC-based security frameworks improved accountability in distributed environments, reducing downtime caused by mismanaged assets (Kormpakis et al., 2023). Likewise, research in manufacturing enterprises shows that role-controlled AMS provided better oversight in resource allocation and minimized insider-related risks (Marquis, 2024).

Nevertheless, implementation costs remain high, particularly for small-scale industrial players. Systems often require significant technical expertise, limiting adoption in resource-constrained contexts.

2.3.4 AMS in SMEs and Developing Economies

Small and medium-sized enterprises (SMEs) face unique challenges in adopting AMS due to budgetary and infrastructural constraints. Empirical work in African SMEs highlights that many commercial AMS are too expensive or complex to maintain, discouraging long-term adoption (Nwabekee & Ugbaja, 2023). Researchers suggest that simplified role-based models, which require minimal configuration, can make AMS more practical for such organizations. Similarly, Bello, Diyan and Asghar (2025) proposed combining RBAC with zero-trust principles to secure assets in legacy-dominated SMEs, showing improvements in accountability and risk reduction.

These studies emphasize the need for AMS tailored to the realities of smaller organizations, particularly in developing economies where resource wastage and poor accountability are persistent challenges.

2.3.5 Key Lesson for AMS Development

Across education, healthcare, industry, and SMEs, AMS has been shown to enhance accountability, optimize resource utilization, and improve audit readiness. RBAC consistently emerges as the most effective access control mechanism, with newer hybrid models addressing limitations of static frameworks. However, high costs, technical complexity, and interoperability with legacy systems remain unresolved challenges.

These lessons reinforce the need for affordable, scalable, and user-friendly AMS solutions, such as the web-based platform proposed in this study, which integrates RBAC security while adapting to diverse organizational contexts.

2.4 Summary of Literature

The reviewed literature demonstrates that Asset Management Systems (AMS) are increasingly critical for organizations across sectors such as education, healthcare, industry, and SMEs. ICT has transformed AMS into centralized, real-time platforms that improve accountability, enhance utilization, and reduce mismanagement of resources. Security frameworks, particularly Role-Based Access Control (RBAC), remain foundational in ensuring that only authorized users can interact with sensitive functions of the system.

Empirical studies between 2020 and 2025 reveal that while AMS adoption delivers clear benefits, challenges persist. High implementation costs, inadequate ICT infrastructure, and technical complexity hinder adoption in resource-constrained organizations. Additionally, interoperability with legacy systems remains a recurring barrier in healthcare and industrial contexts. These issues highlight the pressing need for affordable and adaptable solutions that can be customized to the realities of developing economies.

The literature therefore justifies the development of Kite Assets, a web-based AMS that integrates RBAC to provide secure, scalable, and cost-effective asset management. By addressing the limitations observed in existing systems, the proposed platform is expected to enhance accountability and improve organizational efficiency.

Table 2:1 Summary of Reviewed Literature

S/N	Author/Year	Topic	Methodology	Drawbacks Identified
1	Nyame & Qin (2020)	RBAC framework for knowledge management systems	Conceptual framework	Limited empirical validation
2	Chatterjee et al. (2020)	Dynamic RBAC in decentralized applications	System design & simulation	Complexity in large-scale adoption
3	Nwabekee & Ugbaja (2023)	Role-based framework for IT access management	Case study	Scalability challenges in SMEs
4	Zaidi et al. (2023)	Blockchain-enabled RBAC for IoT and asset tracking	Prototype design & testing	High computational overhead
5	Saxena & Alam (2023)	Trust-oriented RBAC for secure cloud systems	Experimental framework	Increased latency during verification
6	Marquis (2024)	RBAC to mitigate insider risks in organizations	Empirical case study	Focused only on security, less on cost efficiency

7	Kormpakis et al. (2023)	Digitalization and RBAC in energy sector asset systems	Industry-based analysis	High setup cost, complex integration
8	Bello et al. (2025)	Zero-trust + RBAC for legacy SMEs	Hybrid security framework	Implementation overhead, technical expertise need

CHAPTER THREE
RESEARCH METHODOLOGY

3.1 Research Design

This study adopted the system development research design. The aim of the design was to create a functional solution that addresses the challenges identified in existing asset management practices. System development focuses on building, testing, and refining a technological solution until it performs as expected. It was selected because the purpose of this project was not to collect people's opinions but to produce a working platform that improves the way organizations manage their assets.

The design followed a step-by-step approach. The first step was to understand the nature of the problem by reviewing literature and observing weaknesses in manual methods. The next step involved planning and developing the system features. This was followed by testing each module to confirm that it performed correctly. The final step was to refine the system based on the results from the tests. This process ensured that the system met the objectives set in the study.

3.2 Research Philosophy

This study was guided by the pragmatic research philosophy. Pragmatism focuses on practical solutions and encourages the use of methods that help to achieve useful results. It does not limit the researcher to one strict approach. Instead, it supports the choice of techniques that are most suitable for solving a particular problem. This made pragmatism appropriate for the study because the main goal was to build a system that improves how organizations manage their assets.

Pragmatism allowed this project to combine problem analysis, system design, and implementation without following a single fixed method. The philosophy helped shape the study by keeping attention on what works in practice. It also supported the idea that the value of the research lies in the effectiveness of the finished system. For this reason, each stage of development focused on producing features that address real challenges such as record duplication, delayed updates, and poor traceability.

3.3 System Design

The system design stage focused on understanding how asset management is currently carried out, identifying the major weaknesses in the process, and developing a solution that addresses these problems. This stage provided a clear path from the existing situation to the proposed system. It also helped in organizing the features and structure of the asset management platform so that each component meets a specific need. The system design was guided by the aim of improving efficiency, accountability, and ease of use.

3.3.1 Current System

Most organizations manage their assets through manual procedures such as handwritten registers, spreadsheets, or loosely organized digital files. These methods depend heavily on constant human attention, and records are often spread across different sheets or documents. In many cases, assets are recorded only when they are purchased, but follow-up updates such as movement, repair, reassignment, or retirement are not consistently documented. This makes it difficult to maintain an accurate view of the organization's assets.

Retrieving information from the current system usually requires searching through multiple files. This takes time and increases the chance of error. Records can be misplaced, duplicated, or left outdated when responsibilities change from one staff member to another. Since the process lacks central control, it becomes difficult to determine who last handled an asset or who made specific updates. These weaknesses cause delays and reduce accountability, especially in larger organizations.

3.3.2 Challenges in the Current System

Several challenges were identified in the existing manual or semi-manual approach to asset management. The first challenge is the lack of real time updates. Since records are usually edited manually, changes are often delayed, and the information becomes outdated. This affects decision making and makes it difficult to know the actual condition or location of an asset at any given moment.

Another challenge is duplication of records. When multiple sheets or documents are used, the same asset may appear more than once, or different versions of the same file may contain conflicting information. Human error also plays a major role in this problem. Mistakes made while entering or updating records affect the accuracy of the entire system.

There is also poor accountability. Manual records do not clearly show who last handled an asset or who made specific entries. This makes it difficult to trace responsibility, especially when an asset is missing or damaged. In addition, report generation is slow because information must be gathered from several disconnected files. The lack of a central platform limits the ability of organizations to monitor asset performance, plan replacements, or check asset status during audits.

3.3.3 Proposed System

The proposed system, named Kite Assets, was designed to address the problems identified in the existing asset management process. It is a web based platform that allows organizations to register assets, assign them to users, update their status, and generate reports from one central location. The system is built as a multi-tenant platform, meaning that many organizations can use it independently without affecting one another. Each organization manages its own departments, categories, locations, assets, and users.

The system applies role based access control to ensure that users only perform actions that match their level of responsibility. Staff members can view assets assigned to them. Managers can update asset details and track the condition or location of items. Administrators manage users, catalogues, and all assets within the organization. The platform administrator oversees every organization and can assist with tasks such as resetting an organization’s admin password. This structure improves accountability and reduces the risk of unauthorized changes.

The system introduces features that are not available in manual processes. Each asset has a unique QR code, which makes verification easy during audits. Users can also switch between light and dark themes depending on their preference, which improves the user experience. The dashboards present summaries, charts, and recent activities to give a quick view of the organization’s assets. These features support better monitoring, faster decision making, and improved transparency.

Below are the main design diagrams that describe how the system operates:

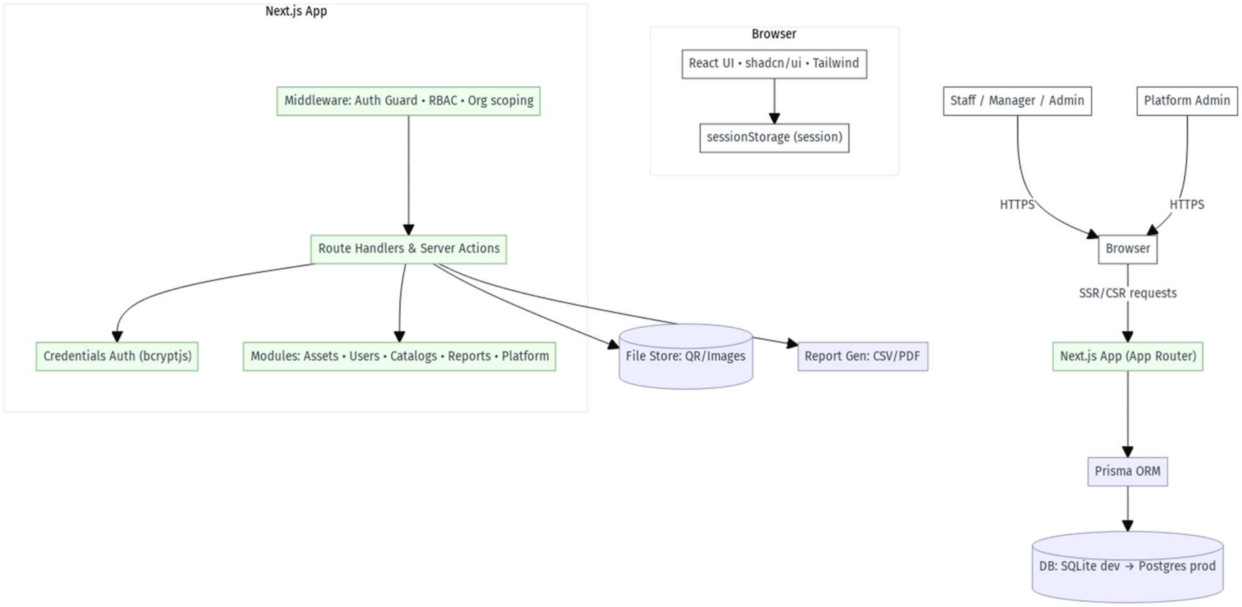

Figure 3:1 System Architecture Diagram (showing client, server, database)

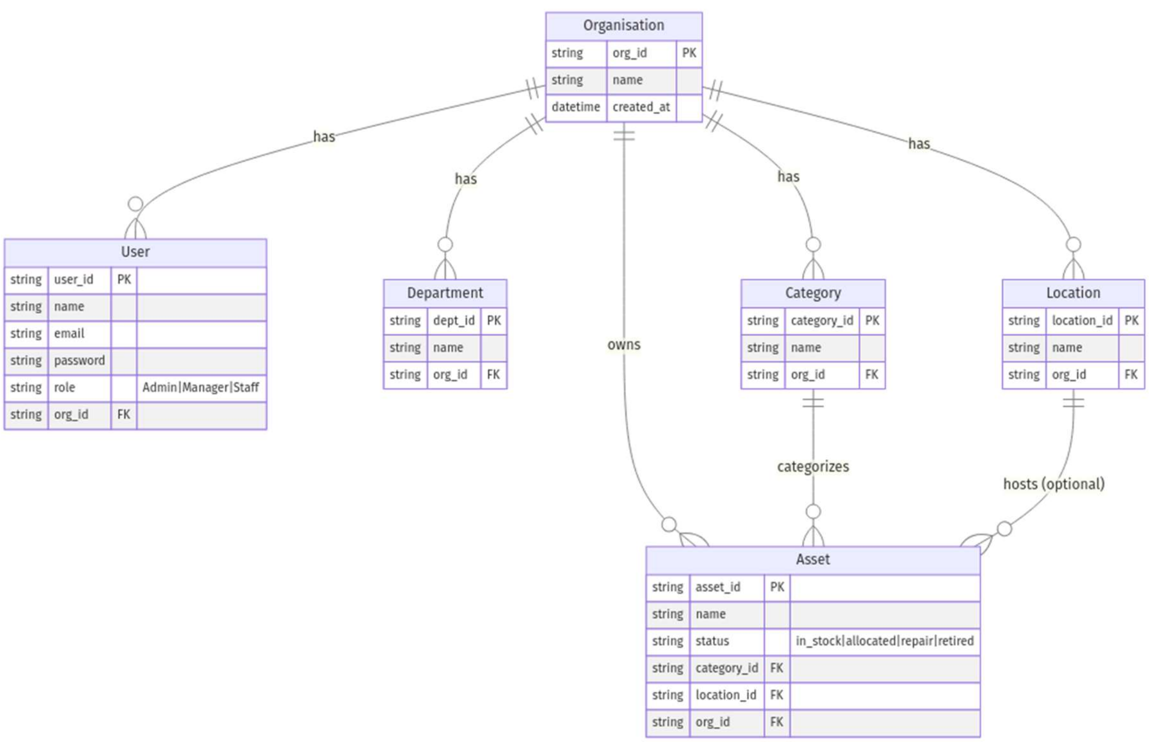

Figure 3:2 Entity Relationship Diagram

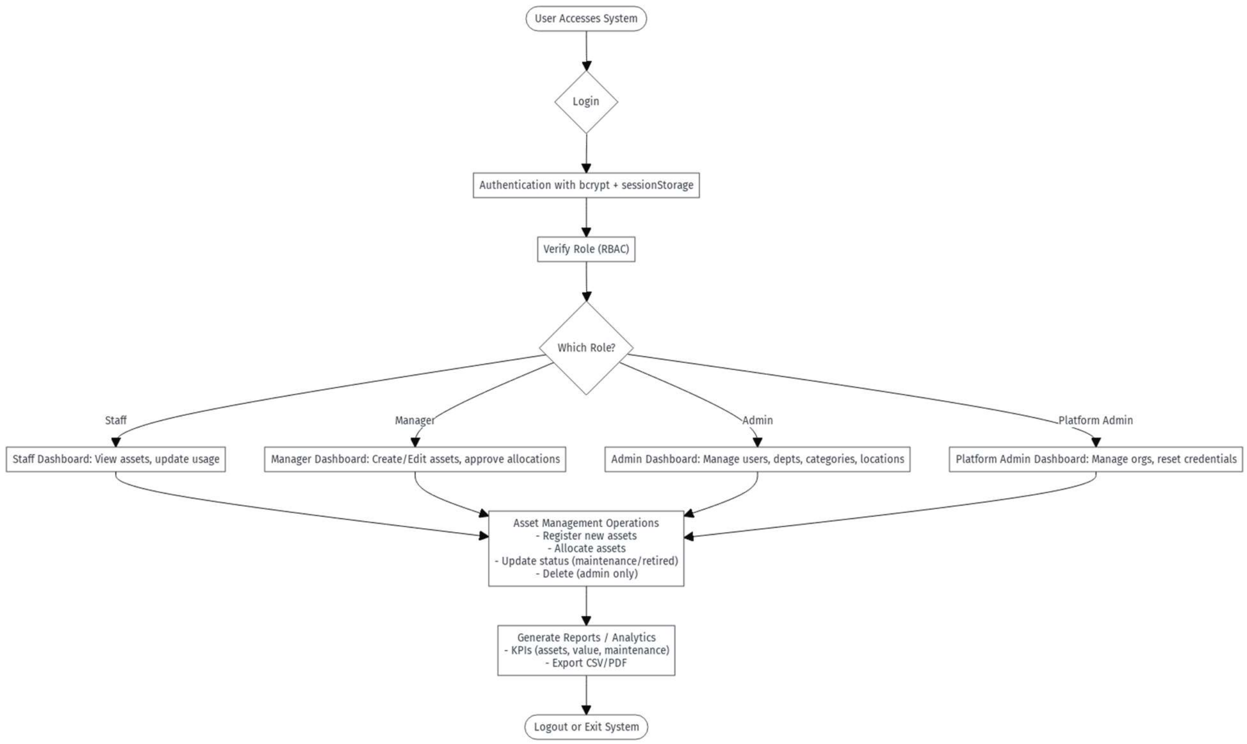

Figure 3:3 System Workflow Diagram

The proposed system was developed using Next.js for the interface and routing, TypeScript for reliability, Prisma ORM for database communication, and Tailwind CSS for styling. These technologies were selected because they support fast development, good performance, and easy maintenance. The design of the system ensures that it can scale as organizations grow.

3.4 Methods of Data Collection

The study relied on three main methods for collecting the information used during system development. The first method was the review of relevant literature. This involved examining academic articles, project reports, and documented practices related to asset management. The review helped in understanding the common problems faced by organizations and the typical features found in modern asset management systems.

The second method involved observing how organizations usually keep asset records. This included noting the spreadsheets, forms, and manual procedures that are commonly used. These observations guided the identification of gaps such as duplication, poor tracking, incomplete updates, and the difficulty involved in generating reports.

The third method was the creation of simulated datasets. Since the system was being developed for general use, sample data was generated to represent users, departments, categories, locations, and different asset statuses. The simulated data helped in testing the system and confirming that each module worked as expected. It also made it possible to test the system repeatedly without depending on sensitive information from any real organization.

3.5 Justification of Methods

The methods used in this study were suitable for the type of system that was developed. The review of literature was necessary because it provided background knowledge on the challenges of manual asset management and the features that make digital systems more effective. Without this foundation, it would have been difficult to design a system that responds to real organizational needs.

The observation of existing practices was also important. It helped in understanding how asset records are handled in everyday situations and revealed the gaps that the proposed system needed to address. This method ensured that the system design was shaped by practical issues rather than assumptions.

The use of simulated data was justified because it allowed the system to be tested in a safe and controlled way. Since real organizations may not be willing to release their asset records for testing, the simulated data provided an alternative that still allowed meaningful evaluation of the system. It made it possible to check each module repeatedly and confirm that the system behaved correctly under different conditions. These methods together supported the aim of building a functional, reliable, and useful asset management system.

CHAPTER FOUR
SYSTEM IMPLEMENTATION AND FINDINGS

4.1 System Implementation

The implementation of the system involved translating the design into a functional web application. The development process was carried out using modern web technologies that support speed, reliability, and ease of maintenance. The system was implemented with Next.js for the user interface and routing, while TypeScript provided type safety and reduced the chance of errors during development. Prisma ORM was used to communicate with the database, and Tailwind CSS was adopted for styling the interface in a clean and consistent manner.

The system was organized into different modules. Each module handled a specific task and was developed independently before being linked with the others. The authentication module handled user login and registration, while the dashboard module presented summaries and quick links. The asset module allowed organizations to add, edit, track, and monitor their assets. User and role management was handled in a separate module to ensure that administrators could oversee staff activities. Additional features such as QR code generation and theme switching were implemented to improve convenience and user experience.

During implementation, each feature was tested immediately after it was developed. This helped in identifying errors early and ensuring that the system worked smoothly. The final product is a web based multi-tenant platform that allows organizations to manage their assets in a structured and accountable manner.

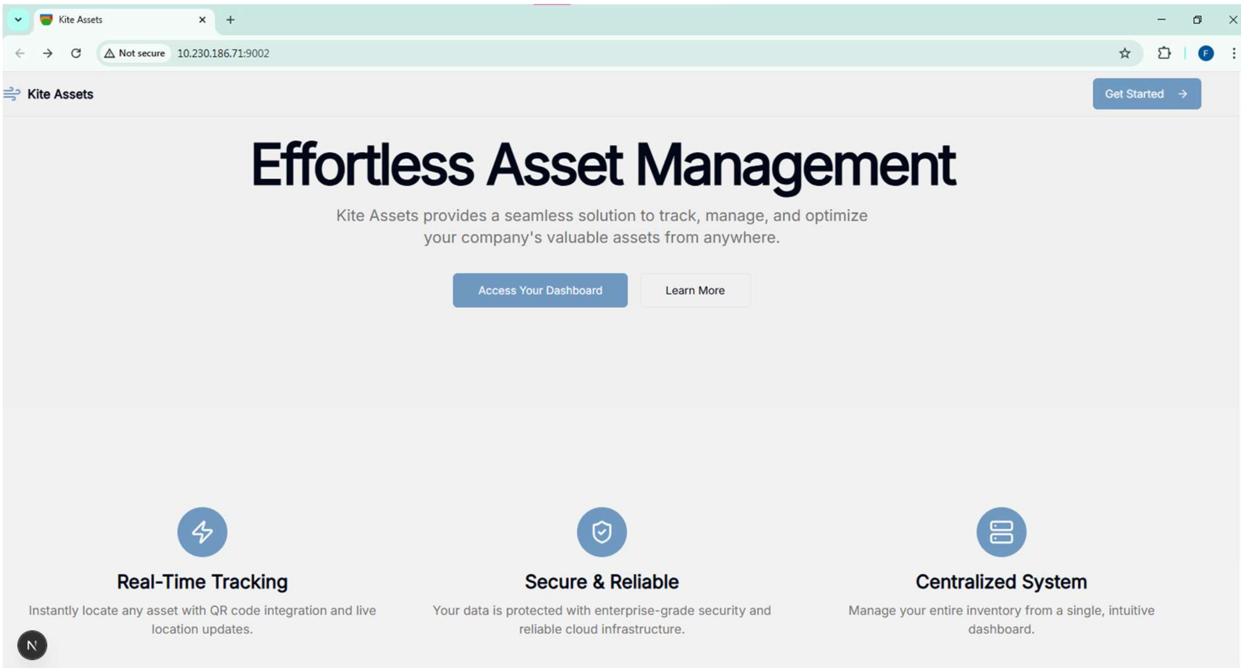

Figure 4:1 Landing Page

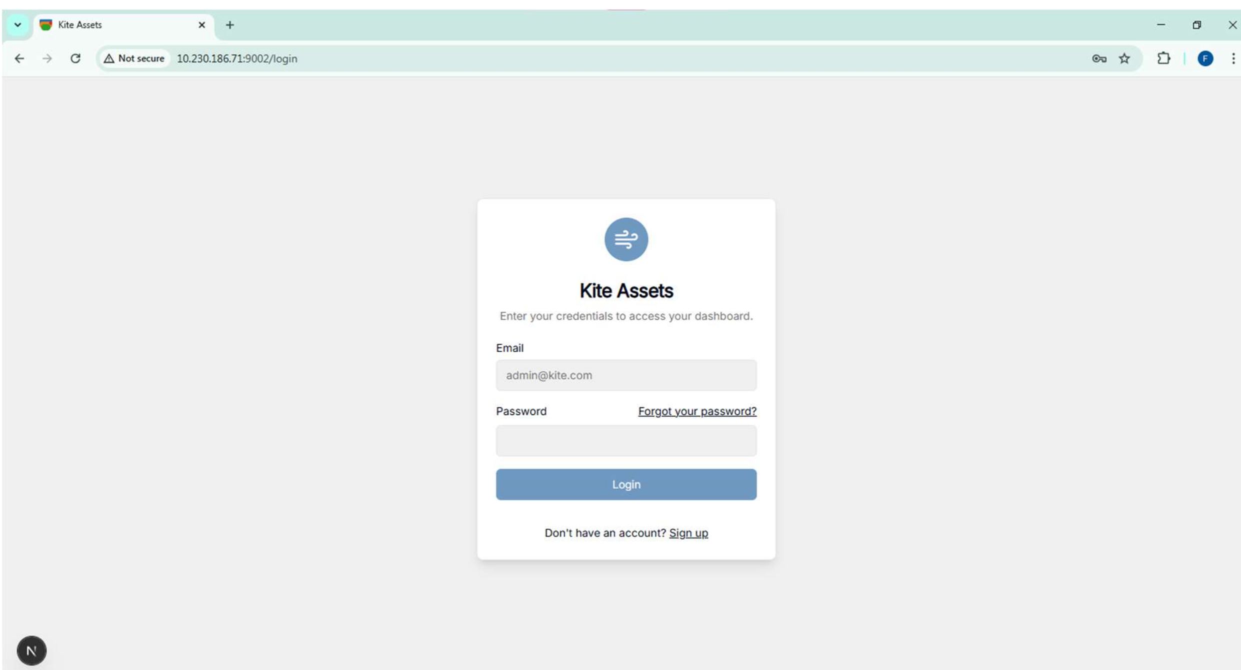

Figure 4:2 Login Page

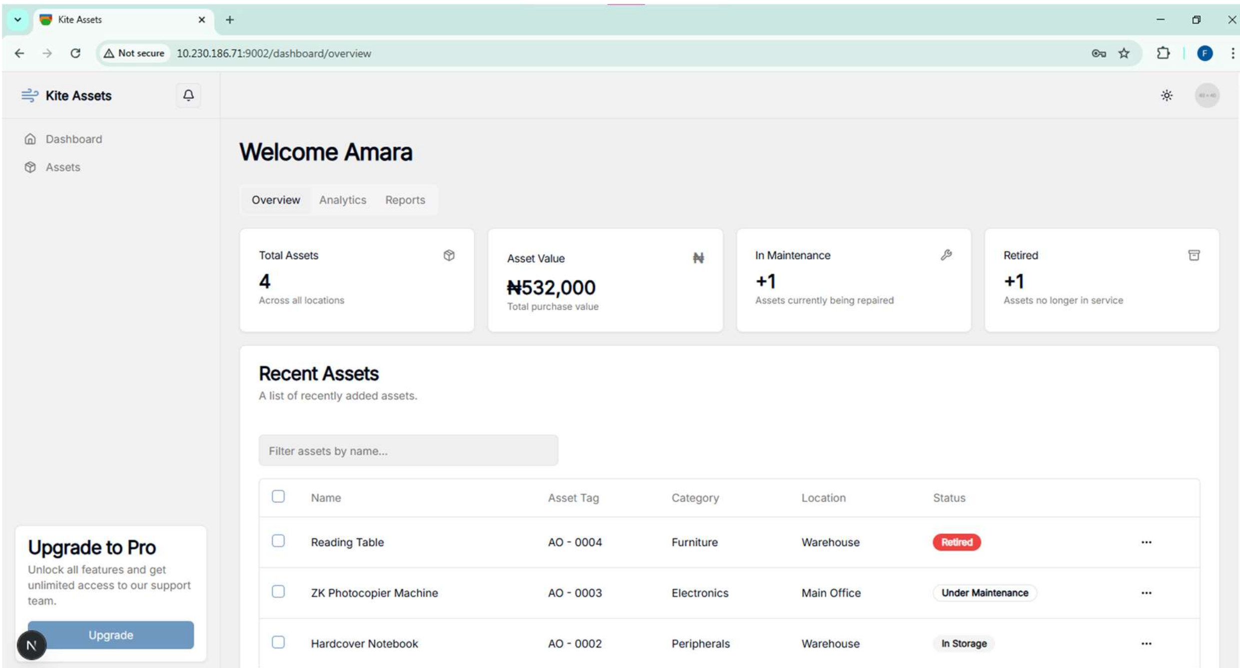

Figure 4:3 Dashboard Overview

4.2 System Requirements

This section describes the requirements needed to develop and run the system successfully. The requirements are grouped into hardware and software components. They outline the minimum specifications necessary to install, test, and deploy the asset management system.

4.2.1 Hardware Requirements

The hardware requirements for the system are modest because it is a web based application. During development and testing, the system ran smoothly on a standard personal computer. The minimum hardware requirements include:

I. A computer with at least 4 GB of RAM

II. A processor equivalent to Intel Core i3 or higher

III. At least 10 GB of free storage space for project files and database

IV. A stable internet connection for installation of dependencies and running development tools

The system can also be accessed by users on laptops, desktops, tablets, or mobile phones. Any device capable of running a modern web browser will support the application.

4.2.2 Software Requirements

The software requirements include the tools and frameworks used in building the system. They also cover the applications needed to run, test, and deploy the platform. The main software requirements are:

I. **Operating System:** Windows, macOS, or Linux
II. **Programming Environment:** Node.js (Version 18 or above)
III. **Framework:** Next.js
IV. **Language:** TypeScript
V. **Database Tools:** Prisma ORM and SQLite for development
VI. **Production Database (Optional):** PostgreSQL or MySQL
VII. **Styling Tools:** Tailwind CSS
VIII. **Package Manager:** npm or yarn
IX. **Code Editor:** Visual Studio Code or any modern IDE
X. **Browser:** Any modern browser such as Chrome, Firefox, or Edge

These tools together supported smooth development and made the system reliable and easy to maintain.

4.3 Discussion of Findings

The development and testing of the system produced several important findings. These findings show how the proposed solution improves asset management and addresses the problems identified in the existing manual process. The features developed during implementation were tested using simulated data, and each module performed as expected. This section presents the behavior of the system and discusses how each part contributes to efficiency, accuracy, and accountability.

4.3.1 Asset Registration and Tracking

One of the major findings is that the system provides a simple and organized method for registering assets. During testing, new assets were created by selecting the category, location, purchase details, and current status. The process was straightforward and did not require any technical knowledge. The system automatically stores each record in the database and makes it available to authorized users.

Once an asset is registered, it becomes part of the organization's inventory and can be tracked across its entire lifecycle. The tracking feature allows managers to update the condition of assets, record movement from one department to another, and change the asset status when it is sent for repair or retired. This creates a complete history for each asset and prevents the information gaps that were common in manual systems.

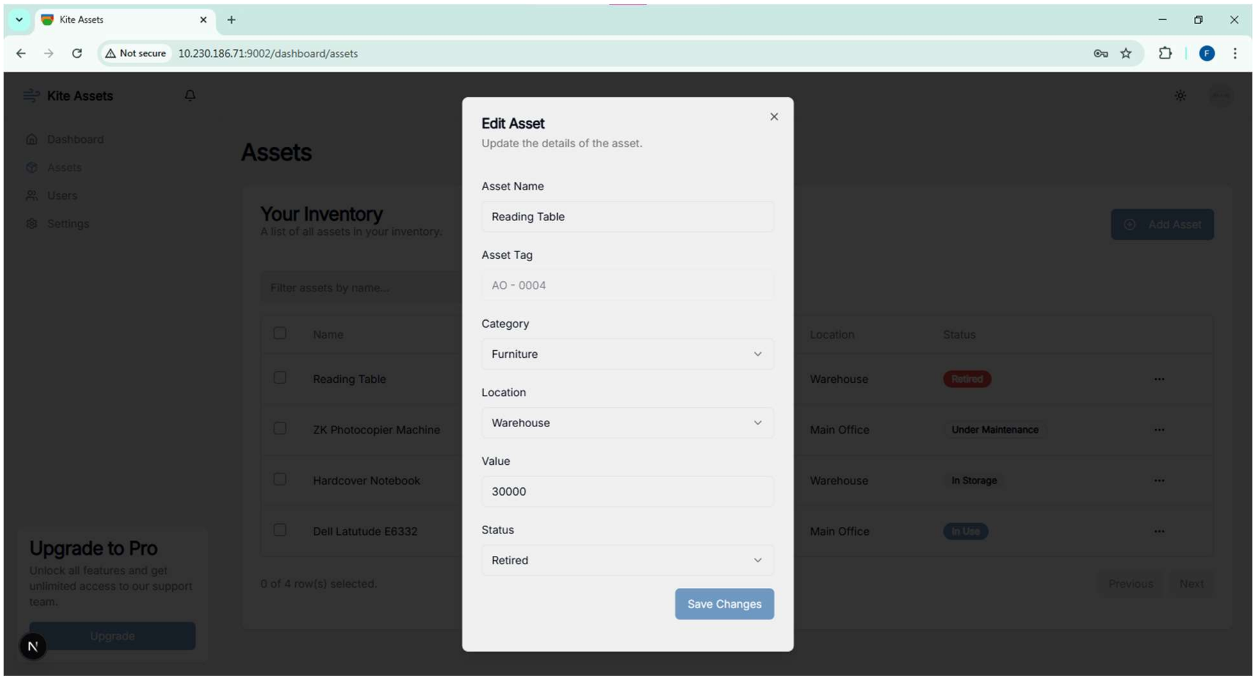

Figure 4:4 Add New Asset Page

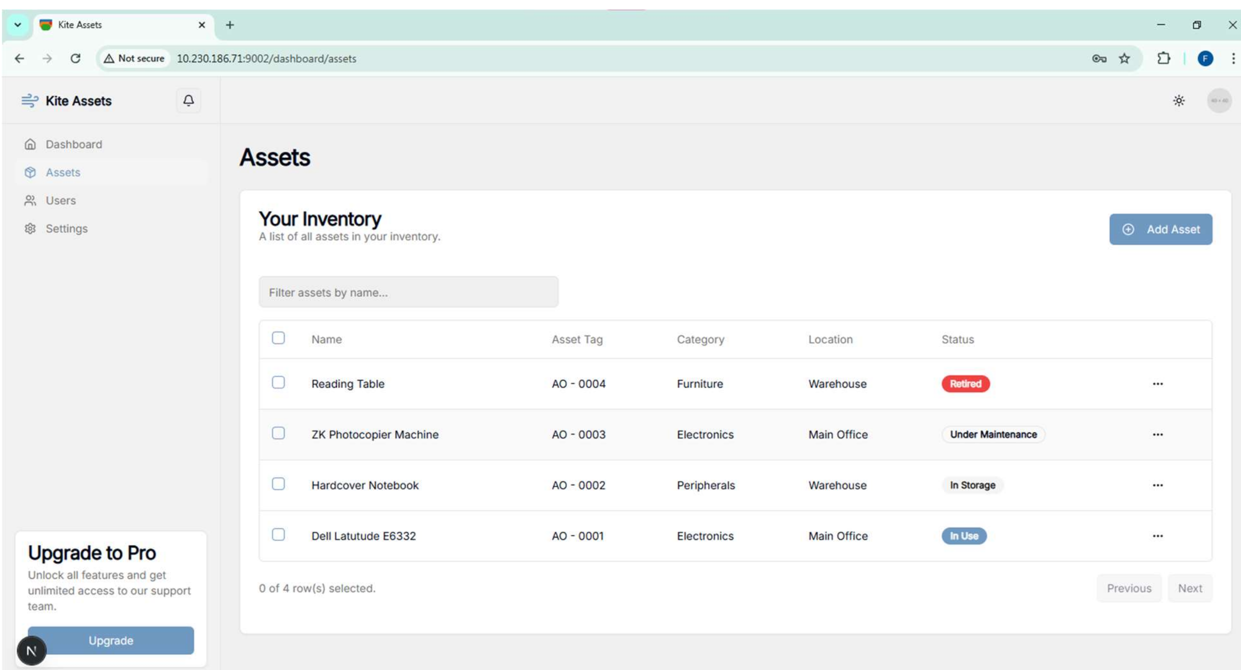

Figure 4:5 Asset Details Page

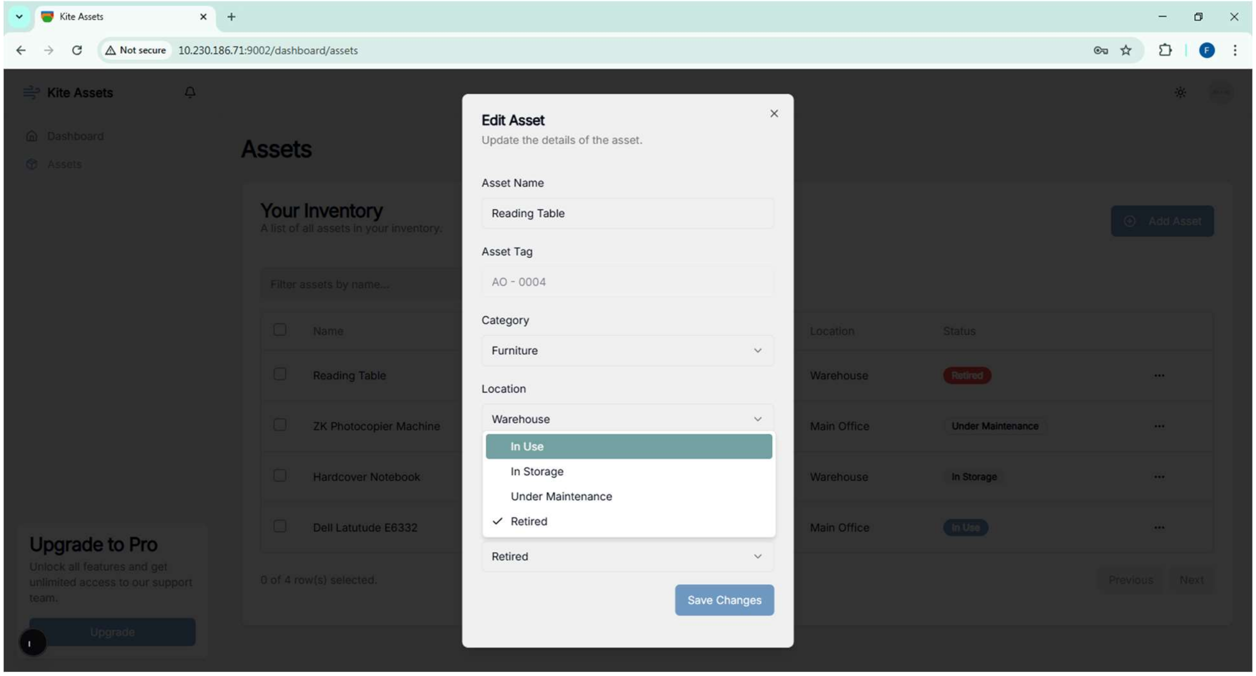

Figure 4:6 Asset Update Form

The findings show that this feature significantly improves accuracy by centralizing all updates in one place. It also increases accountability because each change is tied to a particular user role.

4.3.2 QR Code Asset Verification

Another important finding is the system's support for QR code generation and verification. Each asset automatically receives a unique QR code when it is created. This code contains the asset's identity and allows quick access to its details. During testing, the QR codes were scanned using mobile devices, and the system immediately displayed the correct asset information.

This feature supports fast verification during audits or inspections. Instead of searching through multiple files or scrolling through long lists, a user can simply scan the code attached to the asset. This reduces time and eliminates errors that may arise from manual search.

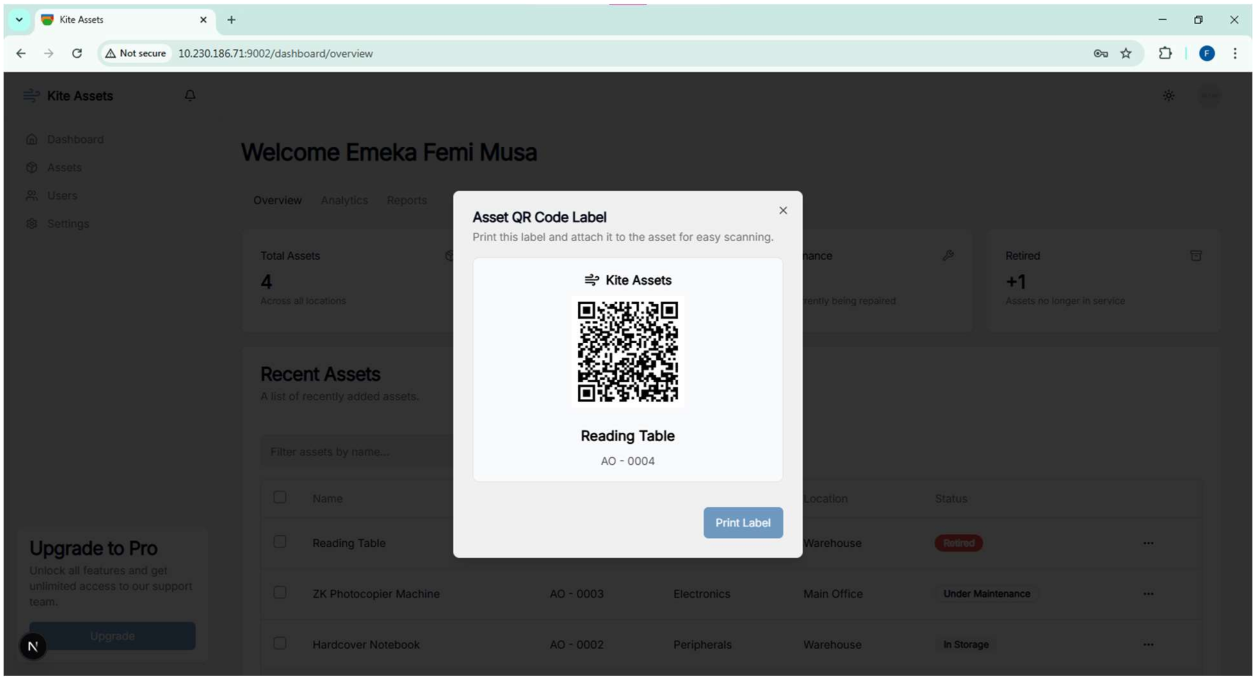

Figure 4:7 Asset QR Code Display

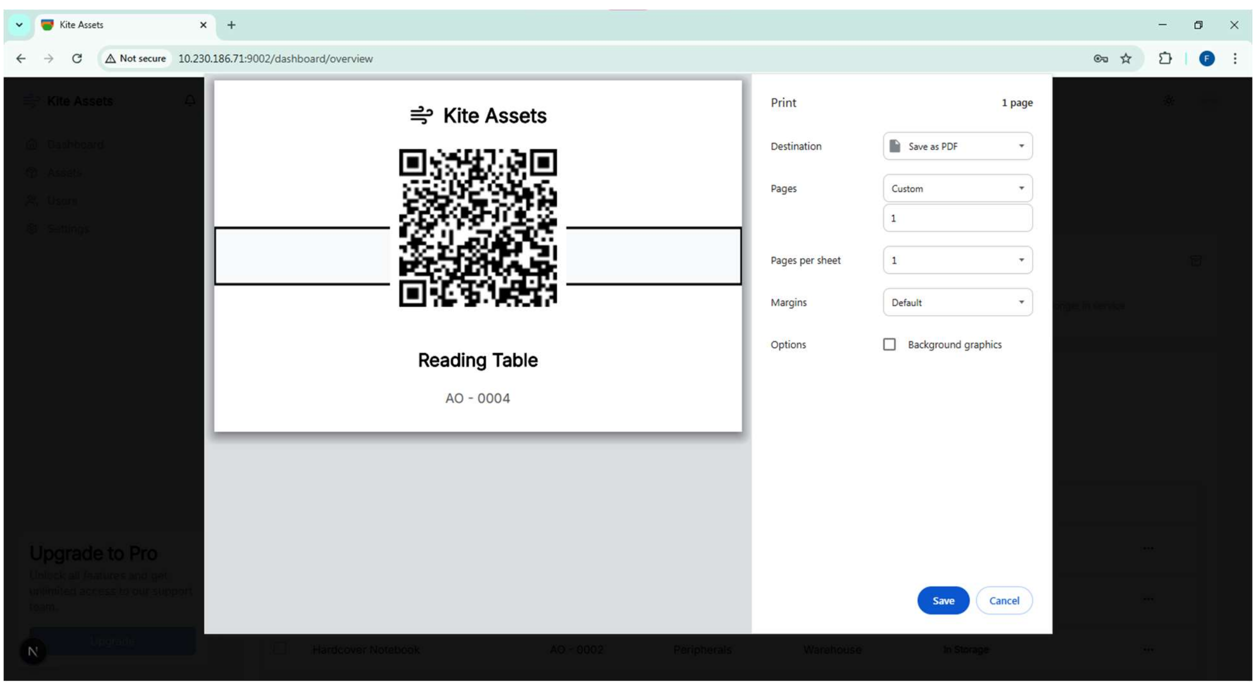

Figure 4:8 QR Code Download/Print Page

The findings show that this feature greatly improves traceability and makes verification more efficient, especially in environments with a large number of assets.

4.3.3 User and Role Management

The system also demonstrated strong control over user access through role based permissions. During testing, staff users were limited to viewing assets assigned to them, while managers could update asset records. Administrators managed users, departments, categories, and other organizational settings. The platform administrator had full visibility over all organizations in the system.

This structure improved accountability by ensuring that every action was linked to a specific role. It also reduced the risk of unauthorized access. The findings confirm that role based access control is effective in maintaining order and protecting sensitive data.

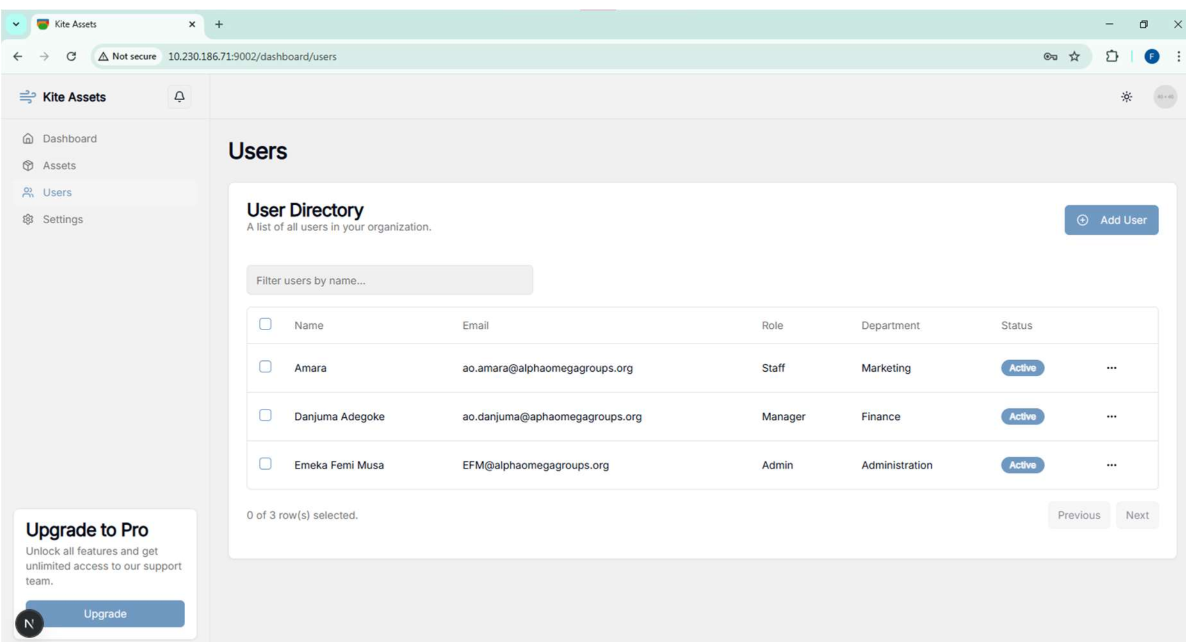

Figure 4:9 User Management Page

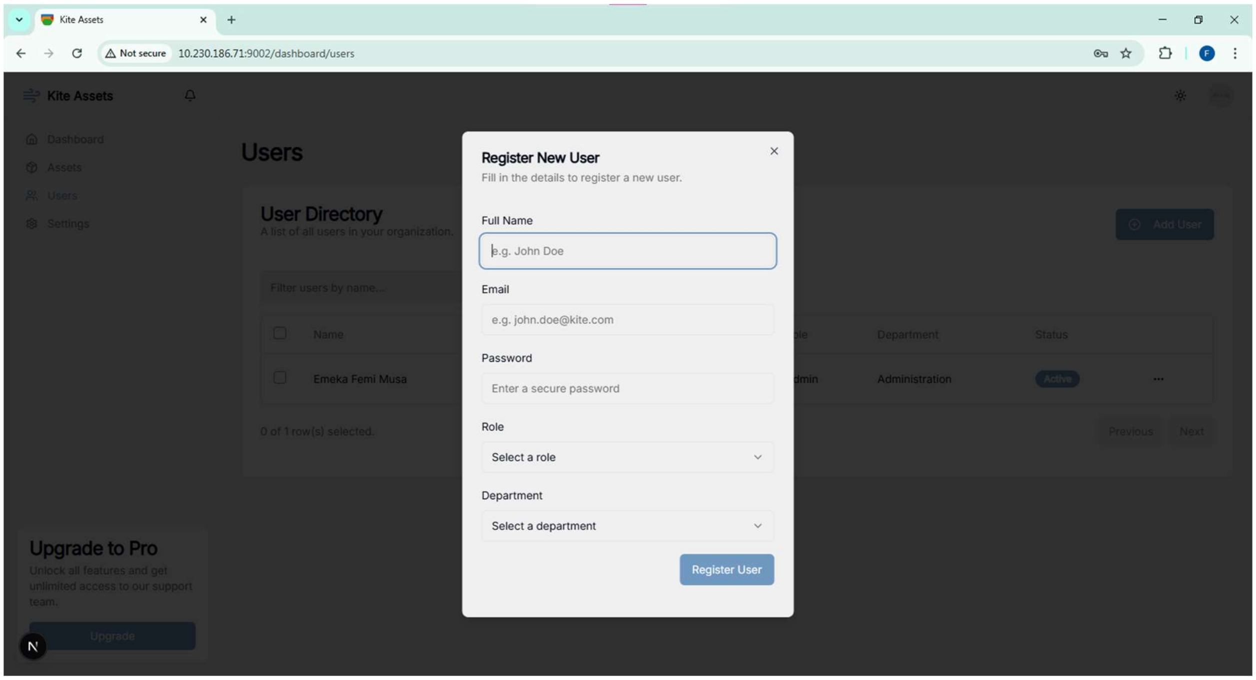

Figure 4:10 Role Assignment Page

4.3.4 Analytics and Reporting

The analytics dashboard provided useful insights during testing. It displayed summaries such as total assets, in-stock items, retired items, and those under repair. Charts on the dashboard helped managers understand the distribution of assets across categories and statuses.

The reporting feature allowed asset lists to be exported in CSV format. This makes it easier for organizations to keep external records, perform offline analysis, or share reports with management.

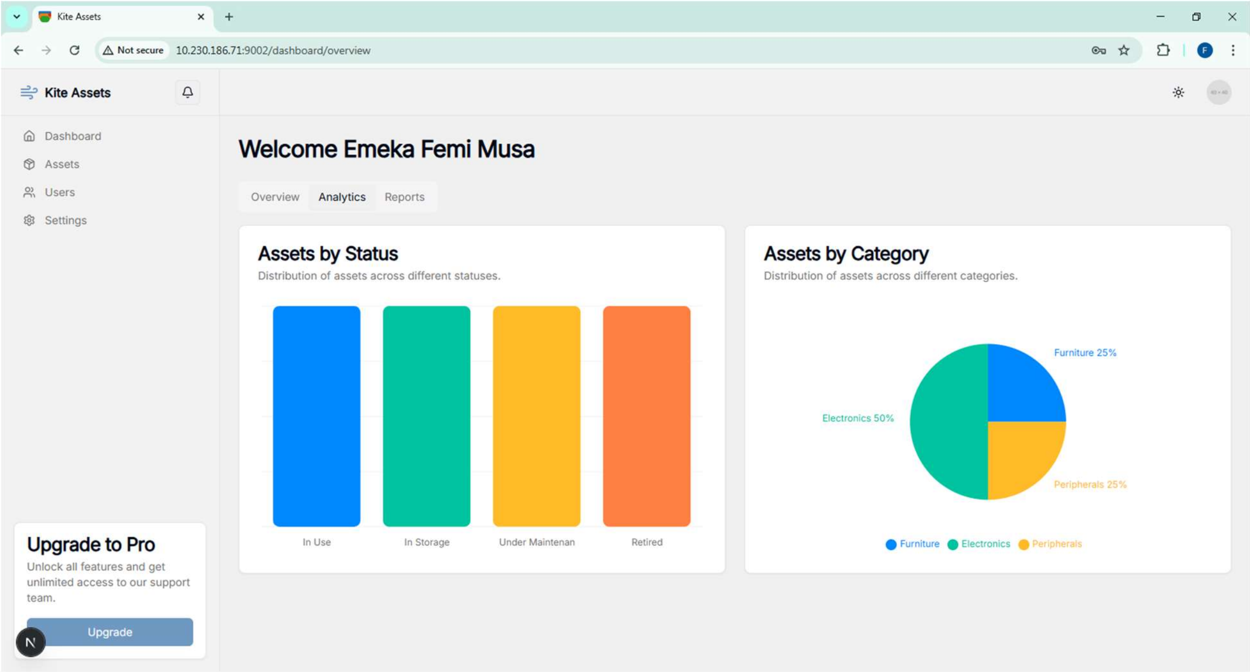

Figure 4:11 Dashboard Analytics Page

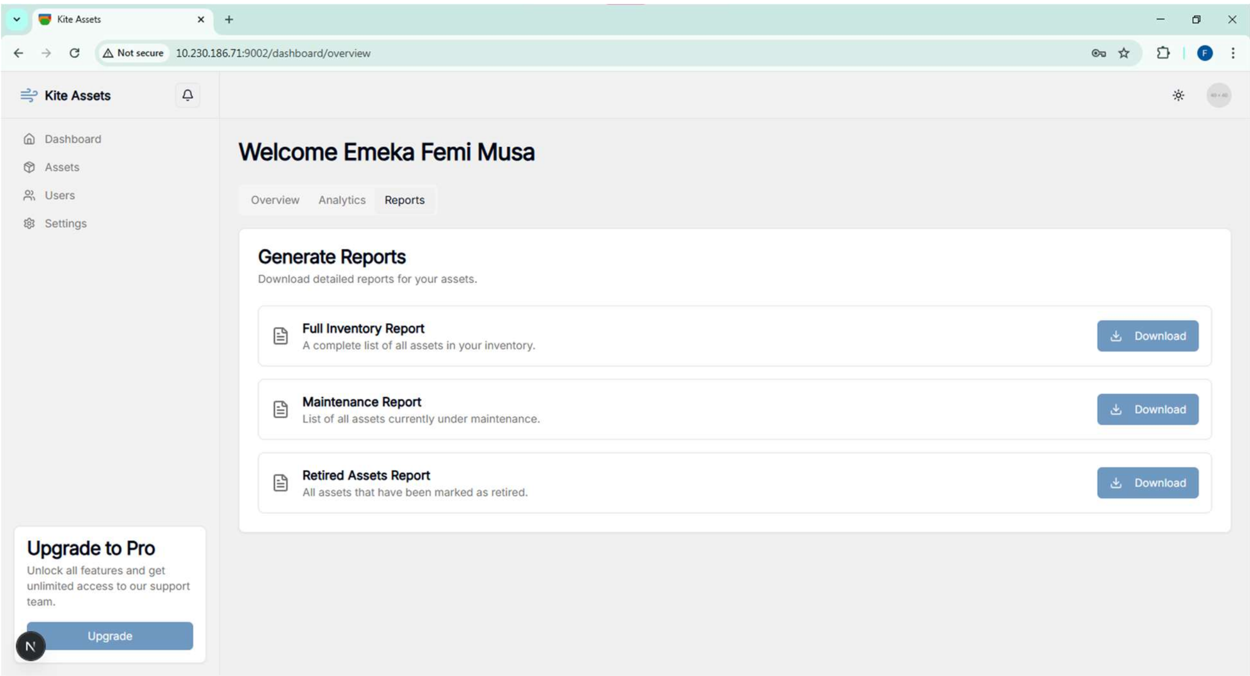

Figure 4:12 Asset Report Export Page

These findings show that the system improves decision making by presenting organised and timely information.

4.3.5 Theme Customization Feature

The system also includes a theme switching feature that allows users to choose between light and dark modes. This feature performed well during testing and changed the appearance of the interface smoothly. It improves user comfort, especially for individuals who work for long hours or prefer a particular visual style.

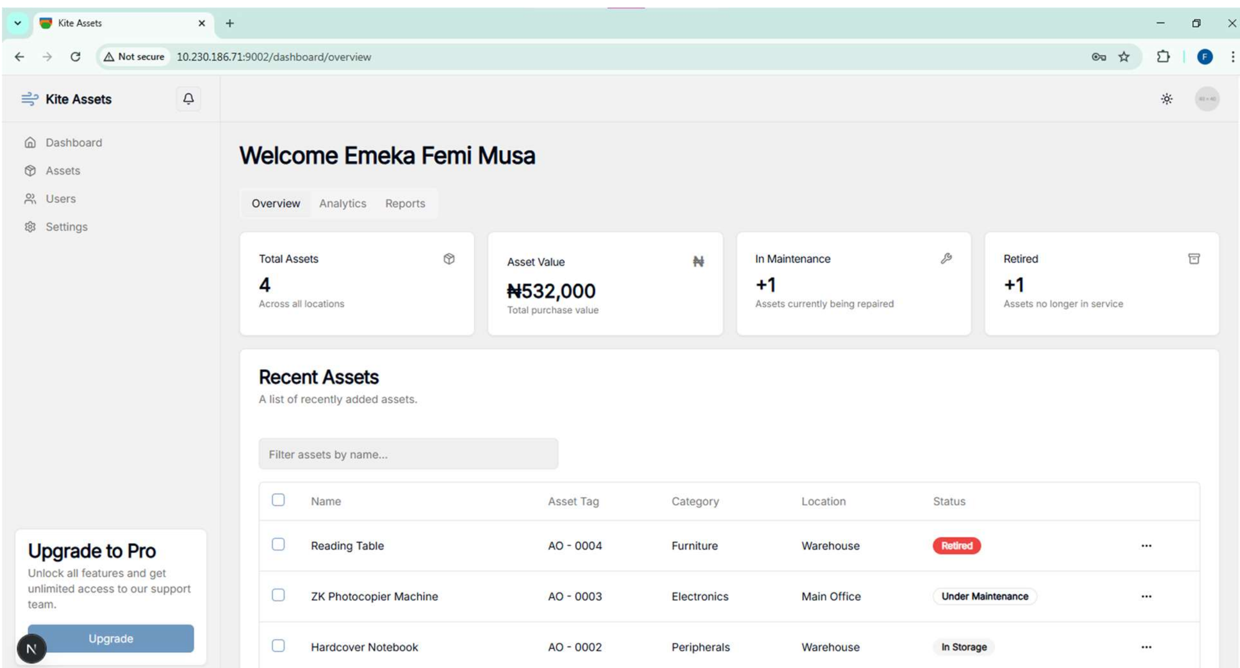

Figure 4:13 Light Mode Dashboard

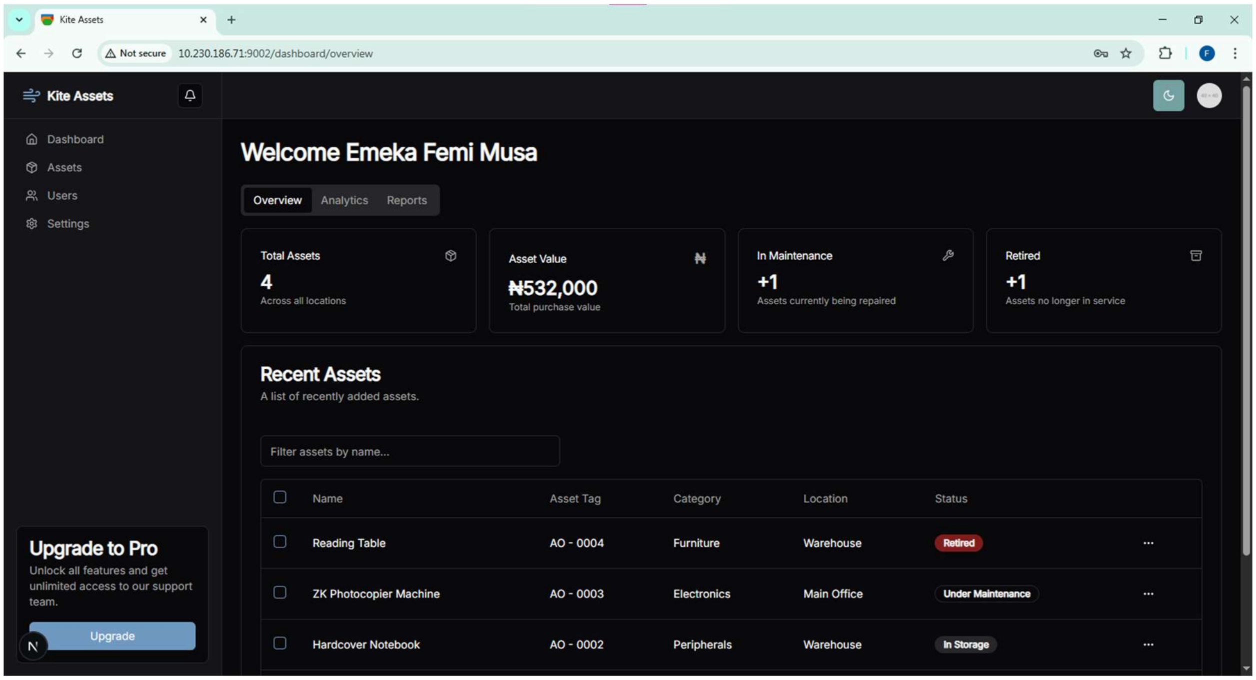

Figure 4:14 Dark Mode Dashboard

The finding demonstrates that a modern and flexible interface increases user satisfaction.

4.3.6 Gap Analysis

This subsection compares the existing manual process with the functionalities provided by the new system.

Table 4:1 Gap Analysis Table

Aspect	Existing Manual Method	Kite Assets System	Gap Closed
Record Keeping	Scattered files and spreadsheets	Centralized database	Eliminates duplication and inconsistency
Updates	Slow and prone to error	Real time updates	Removes delays and errors

Accountability	Hard to trace who updated records	Role based tracking	Improves transparency
Verification	Manual search	QR code scanning	Faster and more reliable
Reporting	Time consuming	One click export	Saves time and improves accuracy
User Experience	Unstructured	Clean interface with theme switching	Better usability

The findings from the gap analysis show that the proposed system addresses all major weaknesses identified in the existing method.

CHAPTER FIVE
SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Summary

This study focused on designing and implementing a web based multi-tenant asset management system. The project began with an examination of the challenges faced by organizations that depend on manual approaches such as spreadsheets and handwritten records. These methods lead to duplication of entries, inaccurate updates, and weak accountability. The study also reviewed literature to understand the features that modern asset management systems should provide.

The system was developed using a system development research design, supported by the pragmatic research philosophy. The design followed a step by step approach that involved problem identification, system planning, implementation, and testing. The final system included modules for user management, asset registration, asset tracking, reporting, QR code verification, and theme switching. The findings from the testing stage showed that the system improved accuracy, speed, and traceability.

5.2 Conclusion

The project achieved its aim by producing a functional and reliable asset management system that responds to the limitations of manual record keeping. The use of role based access control improved the way responsibilities are assigned and monitored. Features such as QR code verification and real time updates strengthened accountability and made audits easier. The system demonstrates that organizations can manage their assets more effectively when supported by a structured digital platform.

The conclusion drawn from this study is that a web based asset management system can significantly reduce the problems associated with manual record keeping. It supports transparency, ensures better organization of information, and provides useful insights for decision making. The system is suitable for schools, small businesses, and other organizations that require a simple but effective solution for asset management.

5.3 Recommendations

Based on the outcome of the study, the following recommendations are made:

1. Organizations that still rely on manual records should adopt a digital asset management system to reduce errors and improve accountability.
2. Administrators should ensure that user roles are assigned properly and updated regularly to maintain clear responsibility.
3. Organizations should establish routine checks using the QR code feature to confirm the condition and location of assets.
4. Further improvements should focus on integrating the system with other applications such as financial or inventory software.
5. Future deployments should use a production level database such as PostgreSQL or MySQL for better scalability.

5.4 Suggestions for Further Study

The study can be expanded in several ways. A mobile application can be developed to allow users to scan QR codes and update asset records directly from their phones. Future work may also include linking the system with Internet of Things devices so that asset conditions can be monitored automatically. Another area for future study is the use of predictive analytics to forecast asset maintenance needs or replacement schedules. These additions would make the system more advanced and improve its usefulness.

5.5 Contribution to Knowledge

This study contributes to knowledge by demonstrating how a multi-tenant asset management system can be built using modern web technologies. It shows that a simple and affordable platform can replace complex or expensive solutions. The inclusion of QR code verification, role based access control, and theme switching distinguishes the system from basic asset registers. The study also adds to the understanding of how digital platforms can improve transparency and management efficiency in organizations.

5.6 Limitations

The system was developed and tested using simulated data instead of real organizational information. This means that some behaviors in real world environments may vary slightly. The development environment also used SQLite, which is suitable for testing but not for large scale deployment. The system currently functions only through a web interface, and no dedicated mobile application was developed. These limitations provide room for future improvement.

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